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Abstinence from social media use, subjective well-being, stress, and loneliness

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Abstract

Purpose: Studies with experimental designs in which the association between participants' subjective well-being and their use of social media is studied remain rare. This study tested the effect of abstaining from social media use on participants' well-being, affect, perceived stress, and sense of perceived loneliness.

Design and Methods: Randomized, controlled design. Sixty-eight participants were randomized to one of two conditions.

Findings: Those who abstained from social media use, compared with those in the control group, evidenced a decline in life satisfaction, an increase in negative affect, and an increase in loneliness.

Practice Implications: Excessive social media use poses deleterious consequences for users, but abstinence may not necessarily exact positive changes; this outcome is dependent on the functions for which social media are used.

KEYWORDS
abstain, loneliness, quit, social media, stress, well-being

1 | INTRODUCTION

Social networking sites (SNSs) have dramatically revolutionized the way in which people communicate and interact with each other. These digital applications allow users to create personalized profiles; share user-generated content with other users; send direct messages to others, both text-based and image-based; and provide commentary on the content of other users.

Over the last decade, the use of SNSs has increased exponentially. The ubiquity of smartphones, in particular, has increased individuals' immediate access to their social media accounts,[1] and a number of mental health concerns have been noted as following from the excessive, problematic use of Internet-mediated communication.[2] In the United States, the proportion of adults who use SNSs increased from 7% in 2005 to 65% in 2015, representing an increase of almost 1000%.[3] Moreover, users of SNSs spend an inordinate amount of time on these platforms. Facebook has recently revealed that its users spend approximately 50 minutes per day using the Facebook and Instagram platforms.[4] In the United Arab Emirates (UAE), Facebook is also the most popular social media platform, with 9 million active users in 2018, which represents a staggering increase of approximately 5.3 million users within the previous 3-year period. In this region of the world, users appear to spend significantly more time per day interacting on SNSs. With the availability of various social media applications, such as WhatsApp, Snapchat, Instagram, and Twitter, in the Middle East, the average daily time spent using social media in this region is now estimated to be 2 hours and 56 minutes.[5]

Given the extraordinary amount of time that people devote to SNSs, a now substantial body of research has emerged that has investigated the reasons motivating the use of SNSs,[6] the personality and psychopathological correlates of excessive SNS use,[7] and the potential consequences SNS use may have for individuals.[8] The totality of this research evidence has converged on the notion that people are motivated to use SNSs to connect with others, to establish new relationships, to maintain existing relationships with friends and family and to browse others' shared content; thus, the primary benefits to be had from SNS use are greater social capital and the maintenance of relationships.[6,9-12]

However, despite these perceived benefits, a burgeoning body of evidence suggests that people experience a range of negative outcomes associated with SNSs. These negative experiences relate primarily to concerns about privacy, SNSs' impact on interpersonal relationships, and the development of problematic use of SNSs. Some individuals have expressed concerns about the lack of privacy and potential misuse of their data that results from the visibility of the activities posted online.[13-15] For example, the viewing of their profiles by employers or family members may result in consequences that directly impact members' occupational lives or personal relationships; some users engage in social comparison in relation to the content they view, which results in the emergence of deleterious body-image issues; and others experience jealousy or envy culminating in disagreements with their romantic partners as a result of their SNS activities.[14] Some studies have found the emergence of "problematic SNS use", which refers to symptoms resulting from excessive and maladaptive SNS use and that are characterized by addictive tendencies, such as irritability when being prevented from logging into social media accounts,[16] a preoccupation with their profiles, and the use of the platform in general (Steers et al., 2016).[17] Ryan et al.'s[16] review suggests that problematic use of SNSs may allow its users a means to avoid the experience of a negative mood state. However, in contradiction, many other studies have found that using SNSs precipitates a commensurate deterioration in a user's mood; moreover, this decrease is usually attributed to the subjective experience of the banality of SNS use and the feeling that one's time has been wasted on a frivolous activity,[10,13,18] Steers et al., 2016.[12] The evidence appears to suggest that excessive or maladaptive use of SNSs is detrimental to users' mood states, and this is further underlined by the plethora of research that links problematic SNS use to diminished levels of subjective well-being.

1.1 | Use of social networking sites and well-being

Cross-sectional studies of the relationship between the use of SNSs and subjective well-being have produced mixed, often contradictory findings, with some studies having found a positive relationship,[19,20] while others have found the inverse.[21-23] Some scholars have postulated that the nature of this relationship may vary depending on a number of different factors, namely, users' intended purpose when using SNSs, the feedback received from other users during their online interactions, and the type of activity engaged in (eg, passive vs active usage). For example, in Rae and Lonborg[24] those participants who used Facebook to engage with and maintain existing social relationships evidenced a positive relationship between Facebook use and levels of subjective well-being, whereas those who used the platform with the express purpose of initiating new relationships manifested a negative associational relationship. Additionally, this relationship was moderated by the feedback participants received from their social connections such that those who received primarily positive feedback experienced increases in subjective well-being, whereas those who received negative feedback evidenced a reduction.[25] A number of studies have found that passive usage of SNSs, in particular, is related to diminished levels of subjective well-being, whereas their active use, especially among females who are hypothesized to be more socially competent and less likely to be drawn into negative online interactions compared with their male peers, correlates positively with subjective well-being.[26-31]

Studies employing longitudinal and experimental designs that have sought to more closely examine the nuances of this relationship have been rare. However, the totality of these investigations suggests that the use of SNSs produces a reduction in subjective well-being[25] and promotes negative mood states[32] over time, particularly following passive use. Conversely, active use of SNSs produced increases to participants' self-reported levels of life satisfaction, but this was not the case for those who engaged in passive usage.[33]

1.2 | Abstinence from social networking sites

Despite the many benefits that SNSs hold, their active use has increasingly been described as an onerous task.[34] Moreover, there is a noticeable trend among social media studies in that they are increasingly seeking to identify users' motivations for limiting their use of SNSs,[13,35] in some cases quitting after periods of sustained use,[15] or for others, the reasons motivating complete abstinence.[11] Similarly, to users of SNSs, those who choose to quit or abstain cite similar concerns, namely, worries about their privacy online, the potential for misuse of their personal information, concerns about the potential for addiction and its impact on their productivity, and ideological objections against social media platforms.[13,15] Moreover, the participants in some studies have expressed a clear aversion towards online communication, instead of expressing a preference for either other activities or face-to-face forms of communication.[11]

Given these findings, prolonged abstinence from social media use may hold a range of potentially positive outcomes, but the potential consequences to users' psychological functioning (ie, affect, well-being, and life satisfaction) resulting from limited usage, quitting, or abstinence have rarely been explored. Thus, we designed an experimental manipulation using a sample of typical social media users and sought to test the following hypotheses:

Hypothesis 1 (H1). *A brief period of abstinence from social media use will improve subjects' experience of well-being (operationalized as self-reported satisfaction with life and affective well-being), relative to a control condition.*

Hypothesis 2 (H2). *A brief period of abstinence from social media use will increase subjects' sense of loneliness, relative to a control condition.*

Hypothesis 3 (H3). *A brief period of abstinence from social media use will reduce subjects' levels of perceived stress, relative to a control condition.*

Hypothesis 4 (H4). *The following variables pertaining to subjects' use of social media (ie, type of engagement with social media (either passive,*

or active use), the number of people followed online, time spent on social media, and frequency of logins) will moderate the effect of abstinence from social media use on subjects' sense of subjective well-being.

2 | METHOD

2.1 | Study design and participants

This study was conducted at a university in the UAE. The study employed a randomized, controlled design. Following the receipt of ethical approval by the social sciences research ethics committee of the institution, recruitment took place from readily available pools of psychology students registered for extra credit, university-associated social media groups, and campus bulletin boards. Following advertisement of the study on campus during the Spring and Summer semesters of 2018, a total of 78 participants (M_{age} = 22.13 years, range, 18-27 years) volunteered to participate. This sample consisted of both men (47.4%) and women (52.6%), all of whom were undergraduate students enrolled in a variety of fields, from faculties including the sciences (33.3%), humanities (35.9%), business (6.4%), law (23.1%), and food and agriculture (1.3%). All participants were citizens of the UAE, were of Arab descent, and identified as practicing Muslims. Participants in the sample were predominantly single (85.9%), resided in the Emirate of Abu Dhabi (57.7%), and had a grade point average that ranged from 1.85 to 4 (M = 2.82, SD = 0.54).

Participants visited the research center and completed paper-and-pencil versions of the baseline assessments (described below). All assessments were administered in English. Despite Arabic being the official language of the UAE, the medium of instruction at the university at which the study was conducted is English. All classes employ curricula based on the American higher education system and are delivered in English, and students are required to demonstrate a high level of proficiency in the English language to gain admission. Thus, we are confident that the assessment measures did not require translation.

Following the completion of the baseline assessments, all participants were randomly assigned to one of two groups by randomly drawing a one (the experimental group) or a two (the control group) from a bag. This procedure was performed by a research assistant who was independent of those who had facilitated the completion of the assessments. Participants assigned to group one were instructed to abstain from all social media use for a period of 7 days, whereas those assigned to group two were instructed to continue using social media as they had usually done. Following randomization, 39 subjects were assigned to the index group and the remaining 39 to the control group.

For those in the abstinence group, in an attempt to ensure adherence to the assigned experimental condition, participants were immediately asked to delete all social media applications from their mobile devices. Because mobile devices are the most frequently used means of social media use for this population, deleting the applications from them would serve to circumvent habitual logging-in behavior. Participants in the index group were also asked postintervention about the number of times they logged into their accounts during the experimental period.

2.2 | Assessment measures

Questionnaires were administered in the paper-and-pencil format while visiting the research center. In addition to a demographic questionnaire in which participants reported their age, sex, current grade point average, field of study, area of residence, and marital status, the following additional assessment measures were administered.

2.2.1 | Social media usage

We asked respondents a number of questions regarding their use of social media. At baseline, all participants reported the weekly amount of time spent using social media (in hours), the number of people 'followed' (or 'friended') on social media platforms, the number of people followed on social media whom they had not met in person, frequency of logins to social media accounts (daily), and frequency measures of active ('How often do you post updates, comment on or like others' posts, post your own photos, videos, or stories?') and passive ('How often do you login to your social media account, but don't post, comment, chat, or send messages to other users?') social media usage, operationalized using an 8-item scale that ranged from 'never' to 'multiple times each day.' At postintervention, participants also reported the extent to which they engaged in face-to-face socializing with others during the experimental period.

2.2.2 | Subjective well-being

The Satisfaction with Life Scale (SWLS[36]) was used to measure current subjective well-being. It is a 5-item scale in which individuals indicate the extent to which they agree or disagree on a 7-point Likert-type scale with items such as *'In most ways my life is close to my ideal'* and *'I am satisfied with my life'*. Higher scores are indicative of a life that is deemed to be more satisfying. The SWLS has been shown to possess multimethod concurrent validity with other well-being constructs,[37,38] including an association with Facebook usage[8] and a negative association with measures of psychopathology.[36,39] Moreover, it has satisfactory test-retest reliability and adequate internal consistency, which typically ranges from .74 to .87.[36,37,39,40] In the current study, Cronbach's α was good and stood at .84.

2.2.3 | Affective well-being

The Positive and Negative Affect Schedule (PANAS;[41]) consists of two 10-item mood scales and was developed to provide a brief measurement of positive affect and negative affect. Respondents rate the extent to which they have experienced each particular emotion (eg, enthusiastic, inspired, irritable, and lonely) within a

specified time period using a 5-point scale ranging from 1 ('very slightly or not at all') to 5 ('very much'). Studies have varied in terms of the timeframe to which the items refer; in the current study, we asked the respondents to refer to their experiences within the previous 3 days. The PANAS has demonstrated sound construct validity, convergent and criterion-related validity, adequate test-retest reliability, and internal consistency.[41-45] For the present study, internal consistency was good for both the positive and negative subscales, with Cronbach's α values of .88 and .89, respectively.

2.2.4 | Loneliness

We used one of the subscales of the Social and Emotional Loneliness Scale for Adults—Short Form (SELSA) DiTommaso et al.[46] The family subscale elicits the respondent's sense of loneliness in relation to familial relationships. Items such as *'My family really cares about me'* and *'I feel close to my family'* are rated on a scale from 1 ('strongly disagree') to 7 ('strongly agree'). The scale has shown satisfactory construct validity, test-retest reliability, and internal consistency in previous applications.[46-48] Internal consistency for the family subscale in the current sample was satisfactory and similar to those previously reported elsewhere for this particular subscale (Cronbach's $\alpha = .76$).

2.2.5 | Perceived stress

The Perceived Stress Scale (PSS;[49]) consists of 14 items that are used to measure how stressful, uncontrollable, unpredictable, and overloaded participants perceive their lives to be. Responses are provided using a 5-point Likert scale ranging from 0 ('never') to 4 ('very often') in response to a stress-related situation experienced during the preceding month. Examples of items on the scale are *'How often have you been upset because of something that happened unexpectedly?'* and *'How often have you felt that you were unable to control the important things in your life?'*; higher overall scores are indicative of subjective perceptions of a more stressful life. The PSS has exhibited convergent validity and discriminant validity, with scores on the PSS having been shown to be significantly correlated with burnout, overall life stress, depression, and anxiety.[50-52] The instrument has also shown adequate test-retest reliability and satisfactory internal consistency, typically ranging from 0.72 to 0.86.[50,51,53] For the current study, Cronbach's α was acceptable ($\alpha = .70$).

2.3 | Data analysis

The (SPSS-17 software, IBM) package was used to compute all descriptive and inferential statistical analyses. Mean scores at the second assessment time point for each primary assessment measure were compared between the index and control groups using an analysis of covariance (ANCOVA), with the baseline scores included in the computation as a covariate to partial out the potential effect these scores may have had on the outcome measure. Effect sizes of the difference between the index and control groups are expressed using partial eta squared (η_p^2).

3 | RESULTS

3.1 | Attrition and compliance

All participants were encouraged to adhere to their assigned experimental condition. The instituted measure in the experimental group of deleting social media applications from their mobile devices ensured that postintervention, no noncompliance was reported. Furthermore, no attrition occurred; 100% of the sample returned to complete the postintervention assessment phase.

3.2 | Descriptive results

Table 1 provides bivariate correlations and descriptive statistics for all measures at baseline for the complete sample. Scores on the

TABLE 1 Descriptive statistics and bivariate correlations for all variables at baseline

	M (SD)	1	2	3	4	5	6	7	8	9	10
1. No. of people followed	411.02 (407.47)	1									
2. No. of daily logins	7.20 (1.99)	−.02	1								
3. Weekly time spent (hours)	7.14 (4.87)	.13	.25*	1							
4. Active use	5.38 (2.47)	.10	.25*	.43**	1						
5. Passive use	6.27 (2.51)	−.21	.54**	.15	.19	1					
6. SWLS	4.83 (1.20)	.16	-.06	.12	.08	−.15	1				
7. PANAS+	3.43 (.84)	.08	−.01	−.01	.01	−.07	.42**	1			
8. PANAS−	2.36 (.86)	.24	−.02	.20	.09	.16	−.23*	−.32**	1		
9. SELSA	5.39 (.93)	.17	−.13	.15	.10	−.25	.60**	.24	−.22	1	
10. PSS	1.81 (.40)	−.07	.05	.10	.07	.16	−.42**	−.21	.40**	−.22	1

Abbreviations: PANAS, Positive and Negative Affect Schedule; PSS, Perceived Stress Scale; SELSA, Social and Emotional Loneliness Scale for Adults; SWLS, Satisfaction with Life Scale.

*$P < .05$.

**$P < .01$.

TABLE 2 Descriptive statistics for all demographic variables at baseline according to the experimental group

	Experimental (n = 39) *M* (SD)	Control (n = 39) *M* (SD)
Age	22.15 (1.98)	22.01 (2.07)
Sex		
Male	20 (51%)	17 (44%)
Female	19 (49%)	22 (56%)
GPA	2.73 (.55)	2.91 (.52)
Marital status		
Single	33 (84.6%)	34 (87%)
Dating	3 (7.7%)	0 (0%)
Married	3 (7.7%)	5 (13%)
Weekly hours on social media	7.38 (5.53)	6.86 (4.05)
Daily logins to social media	6.86 (2.38)	7.62 (1.32)
No. of people followed on social media	360.56 (390.96)	470.17 (425.18)
No. of personal contacts followed	50.53 (66.49)	50.17 (38.23)
Active use of social media	5.37 (2.31)	5.38 (2.69)
Passive use of social media	6.14 (2.61)	6.41 (2.43)

Note: Data are mean (*SD*), or *n* (%).
Abbreviation: GPA, grade point average.

SWLS were positively correlated with the PANAS positive subscale ($r = .42$, $P < .001$) and negatively correlated with both the PANAS negative subscale ($r = -.23$, $P < .05$) and the PSS ($r = -.42$, $P < .001$). Moreover, active use was associated with time spent on social media ($r = .43$, $P < .001$) and passive use with number of logins ($r = .54$, $P < .001$). Table 2 reports descriptive statistics for all the demographic variables stratified for the two experimental groups. As seen, following randomization, the two groups were similar. Using *t* tests and χ^2 analyses, no statistically significant differences were found on any of these variables at baseline (all $P > .05$).

H1: On the SWLS, compared with those in the control group, participants who abstained from social media use reported lower levels of subjective well-being (postintervention mean score of 4.37, SD = 1.26 vs 5.16, SD = 1.14). This difference in postintervention scores, after controlling for baseline scores, was statistically significant ($F(1,75) = 3.84$, $P = .05$, $\eta_p^2 = .05$). On the PANAS, compared with those who continued to use social media as usual, participants who abstained from social media use showed a greater degree of negative affective well-being (postintervention mean score of 2.60, SD = .89 vs 2.36, SD = .84). This difference, after controlling for baseline scores, was significant ($F(1,75) = 5.976$, $P < .05$, $\eta_p^2 = .07$). Examination of the difference between self-reported positive affect, on the other hand, was not statistically significant (postintervention mean score of 3.34, SD = .63 vs 3.31, SD = .81).

H2: When considering scores on the SELSA, participants who abstained from social media use reported substantially more profound experiences of loneliness during the period under examination. The difference in postintervention mean scores was large ($M = 3.81$, SD = 1.30 vs $M = 3.08$, SD = 1.13 for the experimental and control groups, respectively) and, after controlling for baseline scores, statistically significant with a large effect size ($F(1,75) = 8.558$, $P = .005$, $\eta_p^2 = .10$).

H3: On the PSS, participants in the two groups produced scores that were relatively equivalent (postintervention mean score of 1.79, SD = .49 vs 1.78, SD = .51 for the experimental and control groups, respectively) and thus not significantly different after controlling for the baseline scores ($F(1,75) = .15$, $P > .05$, $\eta_p^2 = .002$). Of note, however, despite the nonsignificant result, the experimental group showed a decline in perceived stress between the two assessment points ($M = 1.86$, SD = .42 and $M = 1.79$, SD = .49), whereas those who continued to use social media showed a marginal increase on the PSS ($M = 1.76$, SD = .38 and $M = 1.78$, SD = .51).

H4: We completed a number of additional analyses. First, when the two groups were compared to determine differences in the extent to which they engaged in seeking face-to-face contact with others during the experimental period, those who abstained reported a greater tendency towards face-to-face socialization with others, compared with those who had access to their social media accounts ($M = 7.89$, $SD = 5.31$ vs. $M = 6.61$, $SD = 4.67$), but this difference was not statistically significant ($t(62) = 1.026$, $P > .05$).

Second, based on suggestions in the literature that the effect of experimental manipulations of social media use tends to vary depending on a number of predisposing factors, most notably, the type of typical use of social media (ie, whether active or passive engagement is typical), the number of people followed, frequency of logins to accounts, and amount of time spent perusing social media accounts weekly, we conducted moderation analyses using Hayes'[54] PROCESS macro to determine whether the effect of abstinence from social media on our primary outcome variable (SWLS) would be moderated by any of the aforementioned variables measured at baseline. None of the potential moderator variables demonstrated statistically significant interactions (active use: $b = -.73$, SE = .58, 95% confidence interval [CI]: [−1.88, .43]; passive use: $b = -.11$, SE = .23, 95% CI: [−.58, .35]; number of people followed: $b = -.002$, SE = .001, 95% CI: [−.004, .001]; login frequency: $b = .71$, SE = .73, 95% CI: [−.76, 2.17]; weekly time: $b = -.22$, SE = .12, 95% CI: [−.46, .03]).

4 | DISCUSSION

This study sought to investigate the effect of abstinence from social media use on a number of key psychological variables known to be associated with the use of social media platforms, most notably, subjective well-being. Using an experimental manipulation in which participants were randomized to one of two groups, we found that those who abstained reported a considerable reduction in their satisfaction with life, compared with those who continued to use

social media as usual. Moreover, negative affect increased sharply for those who abstained, as did their self-report of perceived stress, although to a lesser degree.

Although experimental studies such as the present one are sparse, and although most studies of this nature have tended to be correlational, our findings appear to contradict those of previous experimental studies, which have thus far suggested that quitting social media is likely to result in positive changes to well-being and affect[8,55] and that, conversely, its continued use reduces subjective well-being and promotes negative affect.[25,32] The discrepancy here may be the result of the sample's readiness to engage in abstinence. The literature exploring users' motivations for willingly limiting or quitting use of their social media accounts has revealed that, in these cases involving voluntary abstinence, beneficial consequences tend to ensue, such as reduced stress, enhanced positive affect, and an increase in life satisfaction.[13,15,35] In the present study, however, the experimental group engaged in abstinence at the behest of the research team. Participants in the current sample were randomly assigned to abstain: thus, they may not have been ready to abstain and were, therefore, less receptive to this kind of experience. Moreover, the typical factors culminating in people's desire to willingly abstain may have been absent for many of the participants, thus rendering their abstinence behavior during the experimental period involuntary and generally unpleasant experience.

Additionally, if, as the literature suggests, social media is a primary means for social connection, including for the establishment of new relationships and the maintenance of existing relationships,[6,9,10] deprivation of this avenue for gratifying this need would likely be experienced as aversive. The experience of being 'cut off' from one's friends and the fear of missing out on important events and experiences within one's social network may further compound the intolerable nature of this experience. Therefore, an increase in negative affect and a commensurate reduction in well-being can be expected.

A further variable that has received scant attention in the social media literature but that was found to be prevalent in the experimental group in this sample is loneliness. It has often been suggested that contemporary society relies on social media-related interactions to an extraordinary degree for maintaining connections to others, especially among the young adult population.[3] It should then be unsurprising that one of the potential reasons that abstinence from social media engagement is experienced as aversive is that it deprives one of this social connectedness and instead instils a sense of loneliness. In the current study's experimental group, the experience of loneliness elicited by being separated from their most frequent means of connecting with social relationships was substantial and statistically significant when compared with the control group. The experience of loneliness accompanied by the additional cacophony of negative affect and decreased well-being appears to have prompted those in the experimental group to explore an alternate means for connecting with others. A large proportion sought face-to-face socialization with others during the period of electronic deprivation, but this group difference was not significant. Perhaps being accustomed to electronic-mediated interaction meant that those who experienced loneliness were unable to successfully engage in in vivo interactions and obtain an equivalent sense of satisfaction from the interaction. Alternately, after immense reliance on electronic means of social connection, they may now be unskilled at face-to-face interactions and thus continued to report loneliness despite having sought in vivo interactions.

Effects on perceived stress were not statistically significant, but the mean scores between the two groups were in the expected direction at postintervention. Previous studies have similarly found that a period of abstinence from social media use results in a decline in stress and a concurrent decline in well-being.[8,34,56] The present study's findings echo this pattern of results, which at first glance appear at odds with each other. The following hypothesis is offered as an attempt to explain this finding. Scholars have posited that evolution has placed a limitation on the size of the social network that one is capable of successfully managing without difficulty—in this case, the estimation posited is 150.[34] However, given access to social connections offered by social media applications, the potential size of one's social network is exponentially higher than this estimation. Facebook, for example, allows a maximum of 5000 connections. Successfully maintaining relationships in a social network of the size typically afforded by social media can be taxing and exhausting;[9] users often report the experience of social overload[56] or the experience of stress related to social comparisons on social media.[32] Thus, it is likely that a period of disconnection from social media may reduce the stress associated with these demands by one's social network. This is the reason often offered to explain why many people periodically take voluntary breaks from their social media accounts.[8] However, what usually follows this initial reduction in stress is the realization that one's primary means for social connection is absent and because this is a fundamental need for most human beings,[57] loneliness and a reduction in well-being is likely.

Following suit with Verduyn et al.'s[25] recommendations, we conducted a number of moderation analyses. Specifically, we were interested in testing whether the effect on life satisfaction would vary depending on the extent to which participants' engagement with social media was active or passive. Previous research has shown that when users' engagement tends to be passive in nature, they experience lower levels of life satisfaction, and following disconnection from social media, life satisfaction tends to increase.[8,25] This was, however, not the case in the present study. None of the potential moderators tested (active vs passive use, number of logins per day, weekly usage in time, and number of people followed/friends) emerged as significant moderators. A more highly powered study with a larger sample may prove beneficial for retesting these relationships. Additionally, this study was not initially designed for the purpose of conducting moderation analyses; thus, future studies might consider selecting more precise measures of these moderator variables for retesting.

A primary limitation of this study is the use of a convenience sample: a university-aged sample. Individuals interested in the subject matter may have been drawn to participate; however, participants were not aware at the time of recruitment that

abstinence from social media might be required. Moreover, despite a university-aged population being a very specific sample whose experiences may not necessarily be representative of the larger population from which they were drawn, social media use is prevalent in this age range. This study affords an understanding of the psychological consequences of social media abstinence among young adults who regularly engage in social media use. Future studies might consider studying the phenomenon among a more diverse and older sample of individuals.

4.1 | Implications for practice

Social media use can be detrimental to users. Our results reinforce this contention, as participants in the control group exhibited an increase, albeit minimal, in perceived stress, whereas those in the abstinence group showed a decline. However, for practitioners to recommend that users discontinue social media use may not be a wise course of action, as our results also demonstrated that a range of additional aversive consequences may follow from abstinence (eg, increased loneliness and decreased satisfaction with life). Practitioners should identify the specific purposes for which individuals use their social media accounts (eg, maintenance of social relationships). If social media use is problematic and impedes functioning in other areas of the individual's life, a titration of social media use may be recommended but in conjunction with the development of contingency plans that reduce the likelihood of this experience being aversive (eg, encouraging the individual to seek face-to-face social contact if this is the primary purpose for which social media is used and developing a structured plan for how this could be achieved). Importantly, we did not assess whether individuals' social media use was excessive (ie, qualifying as addictive or indicative of pathological levels). Thus, our recommendations may not hold for individuals who evidence clinical levels of social media use.

CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

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